

Richard Lim

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SUMMARY

Rising senior at Bowdoin College majoring in Mathematics and Computer Science and Visiting Researcher at the Naval Postgraduate School. Interested in neuromanifold and optimization geometry with applications to tropical neural networks and representation learning.

EDUCATION

- May 2027 **Bowdoin College**, Brunswick, ME GPA: 4.0/4.0
Bachelor of Arts, Computer Science and Mathematics
Relevant Coursework: Intro Analysis; Statistics (upcoming Fall 2026); Computational Complexity; Multivariable Calculus; Linear Algebra; Artificial Intelligence; Combinatorics and Graph Theory
- Spring 2026 **Eötvös Loránd University (ELTE)**, Budapest, HU With Honors (Kitüntetéssel)
Budapest Semesters in Mathematics
Relevant Coursework: Functional Analysis; Discrete and Convex Geometry; Quantum Logic and Probability (incl. spectral theory and tensor products)

RESEARCH EXPERIENCE

- Length-OOD Generalization Geometry of Tropical Recurrent Networks** June 2026 - Present
Visiting Researcher, Naval Postgraduate School, Monterey, CA
Advisor(s): Dr. Ruriko Yoshida
- Developed a formal learnability framework explaining why length-out-of-distribution generalization is theoretically possible for dynamic-programming targets, grounded in tropical (max-plus) algebra, polyhedral geometry, and optimal substructure.
 - Proved a no-free-lunch transfer theorem for learning function sequences.
- Clustered Mini-batching and Sampling for Stochastic Optimization** August 2025 - Present
Visiting Researcher, HUN-REN SZTAKI AI Lab, Budapest, HU
Advanced Independent Study, Bowdoin College, Brunswick, ME
Advisor(s): Dr. Jeová Farias Sales Rocha Neto, Dr. Bálint Zoltán Daróczy, Dr. András Benczúr
- Devised clustered sampling and mini-batching algorithm to improve final R^2 by $\sim 18.9\%$ over standard random mini-batching.
 - Investigated the interaction between Gradient-Aligned Regression (GAR) and structured batching schemes, including temperature-controlled k-NN and clustering-based batch construction.
 - Geometrically analyzed and empirically demonstrated performance degradation due to gradient cancellation across clusters, contrasting against random batching baselines.
 - Analyzed the geometry of batch-wise gradients to characterize when clustering amplifies versus suppresses GAR's alignment objective.
- Sign2vec: An Expressive Embedding Space for ASL** May 2025 - October 2025
Bowdoin Research Fellow, Bowdoin College, Brunswick, ME
Advisor(s): Dr. Jeová Farias Sales Rocha Neto

- Adapted Spatio-Temporal Graph Convolutional Networks (ST-GCNs) to supervised and self-supervised representation learning, bridging efforts across computer vision and NLP to create a semantic embedding space for ASL.
- Trained and evaluated ML models on >30,000 videos using PyTorch and High Performance Computing, conducting experiments to optimize model performance.

Group Differences in Lexical Search and Retrieval

May 2024 - November 2024

Gibbons Research Fellow, Lexicon Lab, Bowdoin College, Brunswick, ME

Advisor(s): Dr. Abhilasha A. Kumar

- Created a Python-based computational model to study lexical retrieval using data science packages like NumPy and pandas, and NLP tools like the Universal Sentence Encoder (USE).
- Analyzed data from the Semantic Fluency Task (SFT) for >500 participants using the *forager* model, investigating lexical clustering and integrating vector embeddings with behavioral data.
- Wrote E-Prime program for use in electroencephalographic (EEG) studies on lexical retrieval.

PUBLICATIONS

Lim, R., Yoshida, R. (in prep). A Geometric Framework for Length-OOD Generalization on Dynamic Programming Problems.

Lim, R., Neto, J. F. S. R., Daroczy, B. Z., Benczur, A. A. (in prep). k-NAS: k-Neighborhood Aware Sampling for manifold-aware learning.

TALKS AND PRESENTATIONS

INVITED TALKS

Lim, R. (July 2026) Learning Algorithms, Not Answers: A Framework for Length-OOD Generalization on Dynamic Programming Problems. Guest Lecture, Workshop in Operations Research/Systems Analysis. Naval Postgraduate School, Monterey, CA. *Invited by Dr. Ruriko Yoshida.*

Lim, R. (May 2026) k-Neighborhood Aware Sampling (k-NAS) for manifold-aware learning. HUN-REN AI Symposium, Budapest, HU (*Jointly organized with Nanyang Technological University, Singapore*). *Invited by Dr. András Benczúr.*

Lim, R. (March 2026) Clustered Mini-batching and Sampling for Stochastic Neural Network Optimization. Artificial Intelligence Laboratory Seminar on the Manifolds of Machine Learning, HUN-REN Institute for Computer Science and Control (SZTAKI), Budapest, HU. *Invited by Dr. András Benczúr and Dr. Bálint Zoltán Daróczy.*

POSTER PRESENTATIONS

Lim, R., Cooney, A., Hambric, C., Kumar, A.A. (November 2024). Group differences in lexical source use and clustering during memory search. Psychonomic Society 65th Annual Meeting, Psychonomic Society, New York, NY.

Lim, R., Neto, J. F. S. R. (October 2025) Toward an Expressive Semantic Map for Sign Language Processing. President's Summer Research Symposium, Bowdoin College, Brunswick, ME.

Lim, R., Kumar, A.A. (October 2024). CompCog: Modeling search within the mental lexicon. President's Summer Research Symposium, Bowdoin College, Brunswick, ME.

PROJECTS

BrainSpace: A PCA-based Visualization Tool for OOD Generalization

June 2026

A research-forward, easily extensible repository for visualizing neural network training using PCA-based dimensionality reduction to analyze hidden states.

GRANTS AND FUNDING

CXD Funded Internship Grant

\$7000, June 2026

Awarded for the study of Tropical Neural Networks with Dr. Ruriko Yoshida at the Naval Postgraduate School in Monterey, CA

CXD Conference and Networking Mini-Grant

\$910, March 2026

Awarded for the attendance of the HUN-REN AI Symposium 2026 in Budapest, HU

Monthly Mini-Grant

\$1000, December 2025

Awarded for the attendance of AAAI 2026 in Singapore, SG

Bowdoin Research Award

\$5520, May 2025

Awarded for the study of NLP techniques for Sign Language understanding with Dr. Jeova Farias

Monthly Mini-Grant

\$1000, September 2024

Awarded for travel to present at Psychonomics 2024 in New York, NY

Gibbons Summer Research Fellowship

\$5520, May 2024

Awarded for the study of lexical retrieval using computational models with Dr. Abhilasha Kumar

ACADEMIC HONORS

Book Award

2024, 2025

Awarded for carrying a full-course program and receiving a grade point average of 4.0 during the last academic year

Sarah and James Bowdoin Scholar (Dean’s List)

2024, 2025

Awarded to the top 20% of all eligible students with the highest grade point average

TEACHING EXPERIENCE

Teaching Assistantships

Departments of Mathematics and Computer Science, Bowdoin College, Brunswick, ME

Algorithms

Fall 2025

Intermediate Integral Calculus

Fall 2024

Multivariable Calculus

Fall 2025

Linear Algebra

Spring 2025

Intermediate Linear Algebra

Fall 2026

Tutoring

Baldwin Center for Learning and Teaching, Bowdoin College, Brunswick, ME

Fall 2024 - Present

Introduction to Computer Science
Data Structures and Advanced Programming I
Data Structures and Advanced Programming II
Algorithms
Artificial Intelligence
Differential Calculus
Integral Calculus
Intermediate Integral Calculus
Multivariable Calculus
Linear Algebra
Intermediate Linear Algebra
Introduction to Mathematical Reasoning
Data Analysis

Think Teach Academy, Singapore, SG

November 2022 - August 2023

IB Mathematics: Analysis and Approaches Higher Level

IB Mathematics: Analysis and Approaches Standard Level

Cambridge IGCSE Additional Mathematics (0606)

Cambridge IGCSE Mathematics (0607)

LEADERSHIP AND SERVICE

Captain, Bowdoin Quiz Bowl

August 2023 - Present

Managed club budget and led a team of 15 members in weekly practice and intercollegiate tournaments

Resident Assistant

August 2024 - May 2025

Office of Residential Life, Bowdoin College, Brunswick, ME

- Led orientation and community-building activities for 120 first-year and upper class residents
- Resolved conflicts and provided social and emotional support to a floor community of 16

Student Advisor for the Tenure-Track Faculty Search

2024-2025

Department of Computer Science, Bowdoin College, Brunswick, ME

Ammunition Technician

January 2021 - November 2022

Singapore Armed Forces, Singapore, SG

- Planned duty schedules and food rations for the ammunition depot staff of 80
- Developed and executed orientation, and safety and security training programs for posted-in soldiers
- Awarded Best Soldier of the Month and Letter of Commendation from the Battalion CO

MISCELLANEOUS WORK EXPERIENCE

Technical Support Representative

October 2024 - May 2025

Bowdoin College, Brunswick, ME

- Resolved technical issues for staff, faculty, and students in a high-pressure environment at Bowdoin IT

Curriculum Team Leader

November 2022 - August 2023

Ed-XP, Singapore, SG

- Developed and launched education technology products utilizing gamification algorithms
- Trained 10+ interns and led them in curriculum development for IB Mathematics practice tests
- Created internal Excel databases to manage content and optimize workflow

SKILLS

Programming Languages	Python, Java, MATLAB, LaTeX
ML & Data Science	PyTorch, NumPy, scikit-learn, pandas, Slurm HPC
NLP & Speech	HuggingFace, Word Embeddings
Tools	GitHub Copilot, Claude Code, Git, Jupyter, JetBrains IDEs